Title：Explainability and Accountablity in End-to-End Autonomous Driving under Real-World Long-Tail Events

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Abstract: Current end-to-end (E2E) autonomous driving models internalize objective functions that serve as formalized computational optimization targets, together with associated value hierarchies during training. As a result, they operate as unified systems without external observability or controllability. In essence, although E2E models can generate highly accurate actions, they do not satisfy the structural prerequisites for causal inference, thereby creating structural limitations in explainability and liability attribution.

In contrast, purpose in this paper refers to the externalized hierarchy of values and institutional intentions that such objective functions are intended to implement.
This E2E architecture into which a hierarchy of values has been introduced enables the explainability of post-accident analyses essential for regulatory and legal evaluation—including counterfactual inference for unselected tasks, reconstruction of the conditions under which the purpose hierarchy is switched, and verification of value-judgment priorities.

It follows that this paper addresses not only explainability in E2E autonomous driving but also the issue of institutional accountability. As a means toward this end,, this paper integrates medical multifactorial models, causal inference theory, and philosophical accounts of responsibility to reconceptualize explainability in E2E autonomous driving through the lens of causal modeling, with particular emphasis on the transparency of the purpose hierarchy, the feasibility of counterfactual inference, and the development of concrete proposals.

Keywords: accountability; responsibility; liability; explainability; causal modeling; counterfactual inference; long-tail events; autonomous driving

Section-I. Long-tail Events and Social Acceptability

Addressing "long-tail events"—occurrences with low frequency but potentially severe consequences—is a significant challenge in evaluating the safety of autonomous driving systems. Recent advances in simulation technology and generative AI have enabled the reproduction of diverse traffic scenarios in virtual environments (real-to-simulation transfer). However, comprehensively defining and reproducing long-tail events remains difficult and is regarded as one of the primary technical challenges in autonomous driving technology development.

Defining long-tail events requires multidimensional judgment integrating multiple evaluation criteria beyond mere occurrence frequency, such as severity of damage, detectability, and reproducibility. Furthermore, fully simulating complex human driving behavior, traffic environments, sensor noise, and uncertainties in physical interactions is far from straightforward. Therefore, collecting real-world driving data plays an important role in discovering unknown events and validating models. (see Supplementary Explanation No.1)

However, recent advances in technologies like simulation-to-real transfer and domain randomization have demonstrated the effectiveness of simulation-centric development approaches. Therefore, real-world driving data and virtual environment data should not be viewed as mutually exclusive, but rather as complementary approaches.

Furthermore, establishing conditions for long-tail events and formulating safety standards involves not only technical perspectives but also societal acceptability. This makes the development of institutional frameworks, including standardization bodies and regulatory authorities, crucial. Particularly in the certification process for autonomous driving systems, ensuring transparency in safety verification conditions within virtual environments is expected to contribute to enhancing societal trust.

Traffic accident research indicates that driving behaviors and pedestrian actions deviating from average patterns increase accident risk. Consequently, AI models trained solely on statistically average behaviors may not guarantee sufficient safety, necessitating enhanced capabilities to handle abnormal actions and unpredictable situations.

Ref.: The present analysis suggests that Tesla's development strategy does not necessarily contradict the views of commentators who advocate early market deployment on the grounds that "long-tail events should be trained in AI-generated virtual environments."

This may be due to the ongoing technical challenge of defining the conditions that constitute long-tail events within virtual environments.

Section-II. Explainability and Decision-Making

In recent AI research, the explainability of decision-making processes in autonomous driving systems has become a critical theme. Particularly in end-to-end (E2E) autonomous driving models, while high-precision action generation is possible, the interpretability of their internal decision-making structures remains a challenge.
Traditional autonomous driving systems typically structure action plans hierarchically, separating high-level purposes (e.g., ensuring safety, complying with laws) from lower-level control tasks (steering, acceleration/deceleration, etc.).

In contrast, recent deep learning-based models, which rely on deep neural networks (DNNs), tend to implicitly integrate this hierarchy, raising the question of how accountability and explainability can be ensured when accidents occur.

＊Explainability: A technical concept that denotes the technical ability to explain why the model selected a given operation.

＊Accountability: A legal and institutional concept that denotes who bears responsibility when an accident occurs.
(see Supplementary Explanation No.3 Subsection-1)

Depending on traffic conditions, collision avoidance may sometimes be physically impossible. In such cases, decisions aimed at mitigating damage may be reasonable.

Furthermore, decisions like prioritizing legal compliance over protecting human life, or choosing between evasive steering and evasive braking, are highly situation-dependent, making simple prioritization difficult.
(see Supplementary Explanation No.2)

To make such decision-making processes explainable, it is necessary to comprehensively log sensor inputs, risk assessment metrics (e.g., TTC), vehicle dynamics performance (the time required to complete a maneuver based on performance: the time required to complete the maneuver = Time Budget), and alternative action evaluation results. Additionally, storing probability distributions of decisions and evaluation scores can enhance the reproducibility of causal analysis after an accident. Challenges for improving explainability include:

① Developing systematic classification frameworks for long-tail events
② Standardization of objective function design for autonomous driving
③ Standardization of log design to ensure explainability
④ Clarification of transparency requirements within certification systems

These challenges are closely related not only to technological development but also to institutional design.

Section-III. Causal Inference in Accident Analysis

In analyzing autonomous driving accidents, understanding causal relationships is crucial, going beyond merely identifying statistical correlations. From this perspective, causal inference theory provides an effective framework for enhancing explainability.
Causal understanding is generally organized into the following three stages:

Stage-1. Observation Level (Association)
This stage involves identifying statistical patterns from actual accident data.
For example, differences in success rates between evasive steering and evasive braking can

be obtained as observational data.
However, this does not directly indicate causal relationships.

Stage-2. Intervention Level (Intervention)
This stage involves evaluating the effects of specific action choices through controlled intervention or simulation. In autonomous driving, this corresponds to comparing outcomes of different evasive actions through simulations. (see Section-VI. Ref.: NVIDIA)

Stage-3. Counterfactual Level (Counterfactual)
This stage constructs alternative scenarios that did not actually occur to examine accident causes and liability assessments. Digital twin technologies and replay simulations are increasingly utilized to support such counterfactual analysis.
(see Section-IV. Subsection-6)

Integrating this three-tiered structure is expected not only to enhance post-accident explainability but also to contribute to improving safety design. Recent foundation models in multimodal AI and robotics increasingly construct internal latent environmental representations alongside observational data. Systems such as Wayve GAIA-1 exemplify this approach as video-based latent world models. They should not be treated merely as black boxes.
Therefore, research combining causal inference with analysis of internal model representations is expected to become increasingly important.

Section-IV. Pathological Causal Models and Propensity Scores

Subsection-1. Causal Structure Models in Pathology and Factors Contributing to Automobile Accidents

In the fields of medicine and epidemiology, multifactorial causal models are widely used to understand disease onset and progression. It is generally understood that disease onset involves the interaction of multiple risk factors, environmental factors, and protective factors, making it difficult to provide a sufficient explanation using a single-cause model. This analysis introduces the following three classifications as conceptual tools for organization:

＊Generator: Factors directly triggering disease onset
(e.g., pathogen infection, genetic mutation)
＊Promoters: Factors that accelerate disease progression or increase risk
(e.g., lifestyle habits, environmental stress)
＊Inhibitors: Factors that suppress onset or progression
(e.g., immune function, preventive medicine)

However, these three categories should be understood as a conceptual framework for organizing complex causal relationships rather than a strict standard model in medical research. Disease onset typically arises from the interaction of multiple factors and is rarely explained by a single factor alone. This multifactorial structure may share important similarities with the complexity of accident causation in autonomous driving systems.

Subsection-2. The Statistical Framework and Role of Propensity Score for Causal Inference

When estimating causal effects from observational data, "propensity scores" are widely used as a method to control for confounding effects. A propensity score is an estimate of the probability that a subject receives a specific intervention given observed covariates. Within this framework, propensity scores serve the following roles:

Role-1: Modeling background factors that influence intervention assignment as covariates
Role-2: Enhancing comparability between the intervention and control groups
Role-3: Reducing bias in the estimation of intervention effects.

It is crucial to note that propensity scores do not explain biological causal mechanisms themselves; they are a statistical adjustment method for observational studies.
In this regard, pathological causal models and statistical causal inference methods serve distinct roles.

Subsection-3. Conceptual Linkage and "Role Clarification" Between Pathological Causal Models and "Propensity Scores"

The relationship between the two can be organized as follows:

* Generators may be included as background factors influencing intervention selection or risk assessment within the covariates of a propensity score.
* Promoters may affect both the outcome and intervention assignment and are typically treated as confounding factors.
* Inhibitors may be treated as interventions themselves or modeled as protective covariates.

This framework clarifies the division of roles between "Biological causal understanding" and "Statistical causal inference methods".

Subsection-4. Conceptual Application to Autonomous Driving Systems

The causal perspective derived from pathology may be applied to the explainability analysis of autonomous driving accidents. In autonomous driving:

* Hazard triggers (e.g., vehicles decelerating abruptly, pedestrians darting out) correspond to "Generator factors",
* Environmental conditions and vehicle states act as "Promoter factors",
* And intervention operations such as braking (or in some cases acceleration), steering, and evasion decisions serve as "Inhibitor factors".

It should be emphasized that this correspondence is purely analogical and does not constitute a direct application of medical causal models.

Subsection-5. Causal Modeling of Purpose and Task

Decision-making in autonomous driving can be understood as a hierarchical structure comprising a higher-level purpose and specific control operations (tasks).
Formalizing this structure as a causal model may enhance post-accident explainability. (see Supplementary Explanation No.2)

Specifically:

① Purpose is expressed as a combination of an objective function, constraints, and evaluation metrics (sometimes simply expressed as a loss function or optimization goal).

② Task corresponds to the specific operation selected to realize that purpose.

When the hierarchy of purposes is not made explicit, distinguishing confounding factors from mediating factors becomes difficult, leading to uncertainty in the interpretation of causal inference results. Therefore, making the objective function transparent is not only essential for explainability but also a prerequisite for causal inference.

Subsection-6. Counterfactual Inference and Post-Accident Explainability

In post-accident analysis, counterfactual comparison between the action actually taken and plausible alternative actions is crucial. Recent research has explored counterfactual evaluation utilizing:

1. Simulation-based reproduction technologies,
2. Digital twin systems
3. Causal graph analysis.

These approaches may contribute to the quantitative assessment of accident causation and to improvements in safety design. However, they differ fundamentally from causal reasoning at the moment of decision making.

Subsection-7. Summary of Section-IV

"Pathological causal models" provide a useful conceptual framework for understanding complex causal chains, while "propensity scores" offer a statistical method for estimating intervention effects from observational data. (see Supplementary Explanation No.5, No.8)
By conceptually integrating these two approaches, organizing the Purpose|Task structure in autonomous driving as a causal model could contribute to systematizing post-accident explainability and liability analysis.

These proposals help clarify the structural requirements for future explainability and liability frameworks. Their further development can serve as a catalyst for building more coherent and accountable autonomous-driving governance.

Section-V. AI Model Parameter Disclosure and Causal Explainability

In recent years, the disclosure of trained parameters of large-scale AI models as so-called "open weights" has increased. This facilitates researchers and developers in verifying model architectures, retraining, and applying models, and holds certain significance from the perspectives of reproducibility and transparency.

However, model parameters primarily reflect statistical correlation structures extracted from observational data and do not necessarily represent causal relationships. Recent research has explored the incorporation of causal representations, such as causal world models, into large-scale AI systems. This aligns with the scientific methodological principle that "correlation does not necessarily imply causation."

Explainable AI (XAI) has evolved as a technology to visualize the internal reasoning behind models, but it primarily focuses on interpreting statistical associations.
In contrast, AI based on causal inference (Causal AI) differs in that it aims to understand causal structures through estimating intervention effects and counterfactual reasoning.

Therefore, while publishing open weights contributes to transparency, it alone does not guarantee causal validity. To use AI model outputs as scientific evidence, it is necessary to combine approaches such as explicit causal modeling, verifiable hypothesis setting, and counterfactual analysis.

This perspective also offers important implications for the explainability of autonomous driving systems. This is because, beyond mere prediction accuracy, post-accident verification requires understanding and explaining the causal structures underpinning the system's action selections.

Even if model weights capture high co-occurrence frequencies or statistical associations, this does not imply that they encode semantic or logically grounded causal relationships. For this reason, philosophers concerned with ontology and scholars in causal inference maintain the position that weights ≠ scientific evidence that is, model weights alone cannot be regarded as sufficient scientific evidence.

This limitation becomes particularly salient in causal tasks such as counterfactual reasoning and the estimation of intervention effects, where reliance on weight information alone proves fundamentally inadequate.

Section-VI. AI Architecture and Causal Inference

In designing autonomous driving AI, a multi-layered structure encompassing perception, inference, and action generation is common. Organizing this structure from the perspective of causal inference theory enables examination of its correspondence with a three-level model: Association (Observation), Intervention, and Counterfactual reasoning.

* The visual recognition module primarily extracts statistical correlations from observational data, thus occupying a role close to the observation level.
* Conversely, the action generation module functions as an intervention in the environment, while the decision-making algorithm is responsible for predicting the effects of these interventions.

However, in many current autonomous driving AI systems, explicit real-time handling of counterfactual evaluation (estimating outcomes of unchosen actions) remains limited. As counterfactual inference remains technically underdeveloped in many systems, this limitation may constrain explainability. While reinforcement learning and simulation replay techniques enable some level of intervention evaluation, they should be distinguished from rigorous causal inference.
Accordingly, integrating causal inference theory into AI architecture design constitutes a critical direction for future research.

Ref,: According to my own analysis up to the present, NVIDIA VLM backbone (Cosmos Reason) on Alpamayo-R1 architecture, as a virtual-space platform for autonomous driving AI model training, may be characterized as follows:

* The vision module operates primarily at the associational level, extracting statistical patterns from observational data without explicitly representing causal structure.
* The reasoning component does not perform causal inference per se, but instead approximates causal consistency; as discussed in Section-III, it may be interpreted as operating within a form of pseudo-intervention.
* The action module implements executed interventions in the environment but may not systematically support counterfactual reasoning concerning unchosen actions.

Ref,: In recent discussions on explainable AI, a fundamental conceptual issue has emerged.

* The distinction between statistical approximation and causal inference is often overlooked in discussions of explainability.
* Treating attribution-based explanation methods—such as saliency maps and SHAP (SHapley Additive exPlanations)—as sufficient substitutes for causal analysis risks conflating associational (=statistical association) patterns with genuine causal reasoning.
* A central open question in current research is whether association-based explanations can be regarded as equivalent to causal explanations, or whether they are fundamentally different.

The Vision–Reasoning–Action stack in current E2E systems diverges from the three-tier causal model. Vision remains purely associational, Reasoning constitutes only pseudo-intervention, and Action lacks counterfactual evaluation. These discrepancies may impose the structural limits of explainability in contemporary Vision based E2E autonomous driving.

Section-VII. Responsibility and Institutional Challenges

When integrating autonomous driving systems into social institutions, clarifying the concept of responsibility is as essential as ensuring technical explainability.
The act of an AI system selecting driving maneuvers and the identification of responsibility for those selections are fundamentally distinct issues. However, from an existentialist philosophical perspective, action selection and responsibility are considered inseparable (see Supplementary Explanation No.3).

However, because AI lacks self-awareness and moral agency, responsibility must be distributed among multiple actors: designers, operators, manufacturers, and institutional designers. Therefore, the issue of responsibility for autonomous driving accidents cannot be resolved by technical explainability alone; coordination with "institutional design" is indispensable.

Furthermore, since the decision-making of autonomous driving systems is a collective product of multiple factors, including algorithm design, data selection, and operational environments, it is difficult to attribute responsibility to a single entity. This structural complexity generates institutional challenges that differ fundamentally from those associated with conventional human driving.

The explainability of autonomous driving should therefore be understood not merely as a technical problem, but as a multidimensional issue spanning causal inference, institutional design, and philosophical inquiry.

Section Conclusions

Explainability in autonomous driving is not confined to the visualization of logs or the internal readability of models. The essential questions are: "Why was that action selected?", "Under which purpose was that selection made?", and "Who determined the priority of that purpose, and through what process?" In other words, the issue lies in the transparency of the hierarchy of purposes and value prioritization.

In addition, it is important to position long-tail events not as mere "exceptional abnormalities," but as unavoidable phenomena inherent in the structure of the world. The real world is nonlinear and multifactorial, and human behavior is not necessarily rational. As a result, even events of extremely low frequency may occur structurally and produce severe consequences. Therefore, the institutional design of autonomous driving should shift from attempting to "eliminate exceptions" toward designing systems on the premise that exceptions will inevitably occur.

Furthermore, this consideration emphasizes the necessity of explicitly modeling the hierarchical structure of Purpose and Task as a causal model. If the objective function is not explicitly specified, post-accident causal inference becomes ambiguous, and the allocation of responsibility remains indeterminate. Unless it is institutionally documented which value was prioritized and at what point the hierarchy of purposes was switched, explainability cannot be established.

As far as current E2E models are concerned, post-accident causal inference is difficult for the following reasons:

1. **Purpose**

- The objective function operates as a black box and cannot be externally inspected or modified.
- The weighting of optimization objectives depends on the training data, making post-training control difficult.

  Note: This includes phenomena such as catastrophic forgetting, where previously learned behaviors degrade as new data shifts the optimization landscape. One contributing mechanism is gradient interference, in which gradients from different tasks conflict and destabilize previously learned representations.

- The conditions and timing under which priorities within the purpose hierarchy are switched (e.g., compliance with the law vs. protection of human life vs. harm minimization, including collision avoidance) are not observable. (see Supplementary Explanation No.2)

  Note: A broader ODD is expected to increase the patterns of switching within the purpose hierarchy.

- The purpose hierarchy varies across highways, urban general streets, rural areas, and other driving contexts.
- The purpose hierarchy also varies at night, in severe weather, during traffic congestion and in situations without traffic-signal control (e.g., power outages).

2. **Task**

A task constitutes an intervention. (see Section- III)

However, current E2E models cannot generate predictions for unselected tasks—that is, counterfactual actions. They cannot produce counterfactual evaluations such as "What if the vehicle had steered instead of braking?" and lack the covariates necessary to construct propensity scores.

If counterfactual inference is not feasible, the allocation of responsibility under the legal system may become indeterminate in the event of an accident.

Note: Current AI models not only lack the capability to estimate the outcomes of unselected tasks, but also face significant challenges in computing counterfactual predictions in real time. One potential measure is the application of high-speed Digital twins. However, when the ODD is wide-ranging, the set of possible task options varies, and predicting outcomes for multiple candidate tasks may impose substantial computational demands.

3. **Fusion of Purpose and Task**

Because purpose (value judgments and reward evaluations) and task (physical interventions) are fused within E2E models, causal separation becomes difficult, thereby limiting explainability.

Note: Value judgments and reward evaluations (such as compliance with the law versus protection of human life) require examination within the domains of law and ethics. If responsibility is grounded in AI-generated value judgments, the locus of responsibility becomes unclear. Therefore, the configuration of the purpose-hierarchy and the conditions under which it is switched should, in my view, be defined through legal and institutional design.

4. **Lack of logs**

To implement a Purpose|Task model, it is necessary to record logs of the following: the purpose-hierarchy; value and reward indicators; physical metrics (e.g., TTC, Time Budget); candidate tasks; and candidate counterfactuals.

Note: Current E2E models typically output only a single action. Physical metrics are computed internally and remain opaque. No logs are maintained regarding switching within the purpose-hierarchy, and evaluation scores for counterfactual tasks are not recorded.

Given the current state of E2E autonomous driving models, significant technical hurdles must still be overcome in order to achieve compliance with international safety standards such as ISO (21448, 26262) and UN Regulations (UN-R). (See Supplementary Explanation No.4 and Proposal)

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The causal hierarchy introduced by Judea Pearl explains relationships among observations, interventions, and counterfactual outcomes. However, institutional responsibility attribution appears to require an additional normative layer that specifies the purpose governing action selection. Because the causal hierarchy itself does not encode such purpose structures, causal inference alone may be insufficient for institutional responsibility attribution.

Supplementary Explanation

No.1: Long-Tail Events & Dual Uncertainty

Long-tail events encountered in autonomous driving may be understood as phenomena arising from the overlap of two types of uncertainty:

＊Ontological Uncertainty

The real-world traffic environment is nonlinear and multifactorial, such that rare but severe events may occur as a structural property of the system. This type of uncertainty cannot be completely eliminated solely through improvements in observational accuracy. Ontological uncertainty refers to structural indeterminacy inherent in complex real-world systems, independent of observational limitations.

＊Epistemic Uncertainty

This form of uncertainty arises from the difficulty of fully capturing environmental states due to limitations in sensor precision, data availability, and modeling capability.
(See Supplementary Explanation.No.3, Subsection-3)

Long-tail events in autonomous driving are therefore considered to emerge from the interaction of these two forms of uncertainty. Rather than treating them as merely exceptional anomalies, it may be necessary to incorporate them into institutional design as phenomena that inevitably occur with a certain probability.

In this context, the following measures may warrant consideration for ensuring safety

1. Restriction of operational conditions
   (such as speed, inter-vehicle distance, time of day, and weather conditions)
2. Driving policy design that prioritizes safety margins
   (For example, when a vision-based autonomous vehicle approaches an intersection while following a tall vehicle such as a large truck, the forward field of view may be restricted. Quantifying this reduction and incorporating it into comparative evaluations of "driving policies" may be necessary.)
3. Standardization of vehicle performance and operating environments
4. Development of an accident analysis infrastructure premised on explainability

These considerations may involve trade-offs with the high mobility efficiency expected from autonomous driving. However, from the perspective of social acceptance, they constitute important issues for examination.

No.2: Separation of Purpose and Task

In the field of automotive accident safety, multiple hierarchical classifications of purposes already exist. The following illustrates one example:

Purpose①: Compliance with traffic regulations > Minimization of accident damage based on protection of human life
(Level A: Collision avoidance > Level B: Damage mitigation)
Purpose②: Compliance with traffic regulations < Minimization of accident damage based on protection of human life
(Level A: Collision avoidance > Level B: Damage mitigation)

Consider a scenario in which the distance to a preceding vehicle suddenly decreases due to its emergency braking. Road surface conditions, tire performance, and vehicle load significantly influence deceleration capacity and braking distance, resulting in substantial variation in individual braking performance.

Note: Highways located near airports tend to suppress nighttime illumination for operational reasons. Under such conditions, immediately after a vehicle accident occurs, the lighting equipment of the crashed vehicle is often no longer illuminated. As a result, following vehicles may recognize the presence of the crashed vehicle with delay—thereby shortening the Time to Collision (TTC)—which can lead to secondary collisions.

Under such circumstances, it must be acknowledged that a rear-end collision may not always be completely avoidable.
Furthermore, traffic regulations operate at multiple normative levels, and questions remain regarding the extent to which deviations from legal compliance may be permissible in order to reduce overall risk.

In light of these considerations, explicitly defining the Purpose | Task structure within autonomous driving systems is essential for ensuring explainability in accident analysis.

When collision avoidance (①-A) is judged to be physically unfeasible, continued braking may still achieve damage mitigation (①-B). In this case, switching the purpose-hierarchy from Level A to Level B allows the rational selection of Purpose Level B.

Alternatively, once collision avoidance under Purpose① is determined to be infeasible, the system may revise the underlying purpose premise itself–from "Compliance with Traffic Laws" (Purpose①) to "Protection of Human Life" (Purpose②). This shift resets the hierarchy from ①-A to ②-A (collision avoidance), potentially justifying the selection of a steering task instead of braking task. For example, the vehicle may attempt to utilize lateral space, such as the oncoming lane or roadside space, to initiate evasive action. (At this point, the steering direction is determined by which of the two is selected.)

However, such steering introduces additional risks. Entering the oncoming lane violates traffic regulations and increases the risk of collision with opposing vehicles. If an oncoming vehicle appears, neither ②-A (collision avoidance) nor ②-B (damage mitigation) may be achievable. Accordingly, even after a change in the purpose premise, the system must continuously reassess the appropriate hierarchy and task selection in context.

From the standpoint of explainability, not only TTC (Time to Collision) derived from sensor data but also the Time Budget—defined as the action time required based on the vehicle's inherent dynamic performance—becomes critical in determining whether a given Purpose level is achievable.

Specifically, the weighting between purposes (e.g., protection of human life versus legal compliance), the prioritization between collision avoidance and damage mitigation, and the timing of switching both purpose premises and hierarchy levels are all central to post-event explainability.

The separation of Purpose and Task thus provides a structured framework for modeling decision-making in autonomous driving systems. Purposes should be defined as higher-level normative objectives—such as protection of human life, legal compliance, and damage minimization—while tasks such as braking, steering, and acceleration and their respective control inputs should be positioned as subordinate operational means.

Formalizing the hierarchical relationships among purposes, as well as the weighting and switching rules between them, is indispensable for ensuring explainability when accidents occur.

Finally, by quantifying TTC and the Time Budget, and by explicitly logging the timing and conditions under which the purpose-hierarchy is switched, transparency in the decision-making process can be significantly enhanced.

No.3: Driving Policy Models & Philosophical Considerations

Subsection-1 : **Accountability & Explainability**

In the materials on autonomous driving that I published in 2025, I stated the following: The key issue with the E2E type, in terms of future societal acceptance including traffic safety and legal systems, is how to ensure the "Driving policy model" provides both explainability (e.g., in the event of an accident) and post-event verifiability.

This is because the E2E type is inherently limited in its ability to explain why a particular operational decision was made, and also in its capacity to reproduce and verify such decisions after the fact. The inherent issue of the E2E type, namely, the lack of explainability and post-event verifiability—is considered to hinder the widespread adoption of SAE Level-4 autonomous driving.

In this context, although the term "explainability" is used in the above passage, I consider that E2E-type autonomous driving systems also present issues related to "accountability."

In general, "explainability" refers, in the context of technical analysis of accidents in automated driving, to causal logs and reproducibility concerning questions such as why the vehicle did not stop, why it did not avoid the hazard, why it selected a particular maneuver, or which inputs influenced the decision.(see Section-II)

By contrast, "accountability" refers, in the context of legal-institutional analysis with respect to safety regulations and related frameworks, to the explainability of responsibility attribution—namely, who bears responsibility, whether responsibility can be legally assigned, and whether institutional sanctions can be applied.(see Section-II)

Based on the above perspective, when analyzing the current situation, it is also a fact that questions such as the following are being raised:

"Could it not be made explainable depending on how logs are designed?"
"Could reproducibility not be ensured through statistical regeneration?"

It is considered that such questions are pointed out because E2E possesses inherent constraints in its internal causal structure, and because recognition has not been sufficiently shared that the responsibility-attribution structure required by institutions is not aligned with the internal causal structure of E2E.

Subsection-2 : **AI Decision-Making and Accountability**

This study adopts the position that entrusting AI with the selection of driving operations (tasks) is fundamentally distinct from granting it responsibility for those selections.

Drawing on the arguments of Jean-Paul Sartre, responsibility is inseparable from freedom in the act of choosing. Although AI systems may possess the capacity to select operational tasks, they lack self-consciousness, moral agency, and legal personhood. Accordingly, technical explainability alone cannot establish the normative legitimacy of decision-making freedom. While NVIDIA's emphasis on "choice and explainability" may be regarded as technological progress, it may not achieve social acceptance unless accompanied by a clearly defined allocation of responsibility and corresponding legal frameworks.

Applying Carl Schmitt's discussion of the "state of exception" and sovereignty to autonomous driving safety, one may regard the agent that determines driving operations in long-tail events—understood as exceptional situations—as analogous to a sovereign authority. From this perspective, the intuition that such an agent should bear responsibility for traffic accidents carries a certain plausibility. However, contemporary legal systems require not only outcome-based evaluation (e.g., the existence of harm and restoration of order) but also ex ante justification (legal grounds, procedures, and accountability) and ex post verification.

Moreover, real-world autonomous driving decisions are collective products of multiple actors, including algorithms, designers, operators, and vehicle owners. Identifying a single sovereign authority is therefore difficult. Consequently, the Schmittian notion of "the mere existence of a decision-maker" must be distinguished from the institutional question of how legal and ethical responsibility should be allocated. Schmitt's proposition—that "sovereign is he who decides on the exception"—concerned the suspension of legal order in emergencies and does not directly provide a doctrine of responsibility.

By contrast, from Sartre's perspective, action selection and responsibility are inseparable. Sartre maintains that the act of choosing constitutes the self and that the individual bears full responsibility for that choice. If choice exists but no subject capable of assuming responsibility is present, responsibility cannot be constituted. Freedom does not precede action as an abstract entitlement; it acquires meaning only through accepting responsibility for consequences. Therefore, if decision-making authority is delegated to AI, it is necessary to determine in advance who assumes responsibility for the resulting outcomes.

When accountability cannot be clearly attributed after an accident, regulatory authorities face heightened political and legal pressure, which in turn encourages defensive approval practices and institutional contraction.

Sartre's notion of "the gaze of the Other" may also illuminate the social acceptance of technology. Only when society—through law, institutions, and users—recognizes that a particular actor bears responsibility can its actions be regarded as legitimate.
If responsibility remains structurally opaque, public trust erodes, and isolated accidents can trigger disproportionate institutional backlash.

Furthermore, if social institutions are understood as structures that impose responsibility regardless of whether the acting subject can genuinely assume it, then AI models may need to be subjected to forms of regulatory sanction.
Absent such mechanisms, widespread deployment of "identical driving policy models" could lead to repeated occurrences of similar long-tail accidents. In such circumstances, AI models may need to be treated as units of institutional regulation, subject to suspension, mandatory modification, or compulsory updates.
Because "identical driving policy models" are deployed at scale, a structural flaw in their decision architecture propagates systemically, making model-level regulation more coherent than case-by-case attribution.

Long-tail events refer to occurrences that are extremely rare yet highly consequential. In probabilistic terms, they reside in the tail of a distribution, far removed from the central tendency (i.e., average or typical situations). I invoke Sartre not merely as a metaphor but as a theoretical foundation, and I contend that delegating operational driving choices without structurally anchoring accountability creates a contradiction between freedom and responsibility, thereby undermining the institutional stability of Level 4 autonomous driving.

Subsection-3 : **The Ontology and Epistemology of Long-Tail Events**

Why, then, do long-tail events occur?

The first reason lies in the nonlinearity and non-rationality of human behavior. Humans do not always act rationally; distraction, emotion, fatigue, and misjudgment can lead to atypical actions. The second reason is the complexity and contingency of the environment. Real-world road conditions cannot be perfectly controlled, and extreme events may arise from the accidental convergence of multiple factors. Although the probability of long-tail events is low, they inevitably occur, and when they do, they can escalate into accidents affecting many people.

At the Copenhagen and Solvay Conferences, Einstein's deterministic critiques were addressed, and the view that "natural phenomena can only be described probabilistically" became institutionalized as the standard interpretation in modern physics. Within this framework, contingency is recognized as an intrinsic part of nature's structure. In response to Einstein's remark, "God does not play dice," Niels Bohr famously replied, "Stop telling God what to do." This exchange symbolically underscores the philosophical acceptance of a contingent world. At the same time, alternative interpretations exist, in which probability reflects epistemic limitations—constraints of knowledge—rather than ontological indeterminacy. Bohr provided philosophical and institutional justification for contingency, while Max Born offered its mathematical formulation, formalizing the principle that the absolute square of the wave function corresponds to probability density.

The well-known anecdote between Albert Einstein and Niels Bohr illustrates a profound difference in philosophical stance. Einstein may be seen as maintaining that contingency arises from the "incompleteness of knowledge," and that, with a complete theory, probability would vanish, implying the possibility of hidden variables.

Baruch Spinoza, whose philosophy deeply resonated with Einstein, viewed the regularity of celestial motion—such as the Earth revolving around the Sun and the Moon around the Earth—as the necessary order of God = Nature (Deus sive Natura), interpreting the universe's lawfulness as an expression of the essence of God. In this sense, Einstein's determinism can be understood as an extension of Spinoza's philosophy.

Furthermore, when considering future AI developments, the seemingly inevitable paradigm of AI scaling laws—dependent on massive volumes of data and substantial capital investment—may likewise be interpreted as a strategy to overcome the epistemic limitations inherent in deterministic frameworks.

Long-tail events that autonomous driving systems must confront in the real world involve an inseparable entanglement of ontological and epistemic factors (See Supplementary Explanation.No.1)

＊Ontological: Properties inherent in the world's structure, existing independently of human knowledge or observation. The real world is nonlinear, multi-body, chaotic, and irregular; "rare but possible extreme events" naturally arise from this structure.

＊Epistemic: Uncertainty arising from the limits of observation, data, and modeling. Because it is impossible to observe or predict all states and interactions in the real world, such phenomena must be treated as long-tail events.

In designing institutions for autonomous driving, the question of how fully to incorporate long-tail events—those that inevitably occur in the real world—stems from a structural tendency to treat them as epistemic, that is, as exceptions arising from the limits of knowledge. However, if long-tail events are understood ontologically—as phenomena that necessarily emerge from the structure of the world—it becomes possible to reconsider and adjust the premises of institutional design.

In other words, given the current inability to fully explain the causal mechanisms underlying operational decision-making in driving, long-tail events should not be treated as "exceptional abnormalities." Rather, autonomous driving policies should be developed on the premise that such events represent ever-present risks that cannot be completely avoided and must be continuously anticipated. Consequently, it is foreseeable that autonomous driving systems will need to accept the premise that achieving mobility efficiency equivalent to human driving is inherently challenging.

Nevertheless, it is essential to institutionalize "driving policy models" in a manner that ensures the maximum possible safety margins. Such measures may include limitations on vehicle speed, minimum following distance requirements, braking performance standards under maximum load, route selection that accounts for effective roadway width and the separation between roadway and sidewalk, restrictions on operating time periods, constraints based on weather and other external environmental conditions, and limitations on lane-departure scenarios.

No.4: Roles and Implications for E2E Autonomous Driving

ISO 21448, SOTIF (Safety of The Intended Functionality), addresses cases in which a system becomes hazardous without any component failure.

In the context of E2E autonomous driving systems, several approaches have been investigated to achieve SOTIF compliance, including:

- Statistical safety assurance,
- Explicit definition of the Operational Design Domain (ODD),
- scenario-based verification,
- Data-coverage analysis,
- Addition of external monitoring safety layers.

In contrast, ISO 26262 deals with hazards caused by failures, such as:

- Hardware failures,
- Random software failures,
- Design faults.

These failure-induced hazards are managed through the Automotive Safety Integrity Level (ASIL). ISO 26262 fundamentally assumes that:

- System requirements are explicitly defined,
- System functions can be decomposed,
- Design, implementation, and verification can be traced in a structured manner.

However, current E2E systems consist of a three-layer structure:

1. Input: sensors
2. Output: lateral steering control (right or left) and longitudinal control (acceleration or deceleration),
3. Internal: AI-based deep neural networks (DNNs).

Given this architecture, it is considered highly challenging for E2E systems to satisfy ASIL-B or higher.

ISO 26262 defines the following safety integrity levels:

- QM : Quality Management level (no safety requirements),
- ASIL A: Minor safety degradation, easily controllable by the driver,
- ASIL B: Moderate safety requirements, demanding strict design and verification processes, including system redundancy and diagnostic functions,
- ASIL C: Potential for severe accidents upon failure, requiring rigorous process management, improved fault-detection coverage, and systematic safety analysis,
- ASIL D: Failures that may directly threaten human life, such as loss of braking or steering at high speed.

Especially ASIL D requires high fault-detection rates, redundant architectures, formal methods, detailed traceability, and stringent verification coverage.

Based on this analysis of ISO standards and current technological trends in autonomous driving, “E2E-based autonomous Taxis” are realistically limited to “Corridor-Type” deployments (See Proposal), such as:

· Geo-fenced Level 4,
· Fixed-route autonomous driving,
· Hub-to-hub autonomy.

In other words, City-wide Level 4 autonomy is unnecessary; the system should be restricted to round-trip operations between major terminals. Such routes are also more viable from a business perspective, for example:

Major Origin–Destination Pairs for Round-Trip Operations

· From Tokyo Station to Narita International Airport or Haneda Airport
· From Tokyo Station to Shibuya Station or Shinjuku Station (or to adjacent hotels)
· From Narita International Airport or Haneda Airport to hotels within central Tokyo
· From Shinkansen terminals to major tourist attractions or reserved hotels

These routes should be reserved for pre-booked trips, and all operations must accommodate elderly individuals, persons with disabilities, and passengers (families or groups) carrying substantial amounts of luggage.

Similarly, for autonomous driving in residential urban areas, the system should not replicate only the main routes of large public buses.

Rather, it should address local community needs, such as: (under periodic round-trip operations)

· From Residential areas to schools, hospitals or shopping malls
· From Residential areas to workplace or adjacent station
· From Residential areas to evacuation facilities or emergency shelter facilities

By limiting the ODD to a Corridor-Type and employing the concept of "accessible designated stops," such systems can be made feasible. Therefore, autonomous driving deployment should begin with ODDs defined as "Lines" and only gradually expand toward "Area"-based operation.

No.5: Fixation on Typical Causality and Exclusion of Atypical Causes

Cases exist in multiple medical and forensic reports in which a skewer-like foreign object penetrated the pharynx and neck during a fall while walking, resulting in fatal outcome due to initial misdiagnosis. The penetration of such objects from the oral cavity into the body is frequently attributed to falls while eating, while the causes of the falls themselves may include contact with street obstacles or collisions with other pedestrians.

However, during the initial clinical assessment:

1. In cases of falls, probabilistic diagnostic reasoning often prioritizes impaired consciousness resulting from head contusion. In other words, heuristics favoring the most

frequent causes in trauma care are applied. Consequently, low-frequency yet potentially lethal atypical injuries are easily excluded from the initial diagnostic model.

2. In reality, a skewer had penetrated from the pharynx to the medulla oblongata. This constitutes an instance of "atypical trauma," which is low-frequency but associated with extremely high mortality. When external bleeding is minimal, such injuries may be overlooked during initial evaluation.
3. Altered consciousness observed during examination is often erroneously attributed to head injury. Notably, medullary injury—which can cause respiratory distress, bradycardia, and reduced levels of consciousness—produces clinical manifestations that closely resemble those of head trauma.
4. In medical malpractice cases, the following points are frequently contested:

- Misidentification of the mechanism of injury
- Failure to detect foreign body penetration
- Anchoring bias in the initial diagnosis (fixation on the first hypothesis based on high-frequency patterns)
- Delay in imaging studies

Considering the causal reasoning in this case, if the medical team had not been fixated on the "typical causal pathway" (fall → head contusion → impaired consciousness) and had instead engaged in counterfactual reasoning–asking, "If we had considered an alternative causal model, would the treatment have differed, and would the outcome have changed?"—it becomes evident that, even with medullary injury, fatal outcome could likely have been avoided through early airway management, foreign body removal, and hemostasis.

The injury mechanism involved "a skewer-like foreign object penetrating the pharynx and neck as a result of the fall," and the lethal pathway followed an atypical sequence:

"skewer penetration → brainstem injury → respiratory arrest → fatal outcome"

Nevertheless, if the medical team omitted this atypical pathway from the causal model and misdiagnosed it as head trauma due to anchoring bias toward high-frequency patterns, the counterfactual causation—"If the skewer penetration had been included in the causal model, medical treatment would have differed, and fatal outcome could have been avoided"—would apply, rendering this case highly susceptible to litigation.
In court cases dealing with comparable matters, what is generally questioned is not the outcome itself, but rather the "foreseeability" and "avoidability" of the event.

In autonomous driving technology, just as in medicine, there is a structural tendency toward "fixation on typical causal patterns" and "exclusion of atypical causes."
In particular, DNN-based perception and decision-making models strongly learn

high-frequency patterns in past data, and as a result, systematically tend to underestimate low-frequency but potentially fatal long-tail events.

This is analogous to the "anchoring on high-frequency trauma patterns" observed in medical practice. Therefore, many serious accidents in autonomous driving tend to occur precisely due to the "overlooking of atypical causal pathways."

For example:

1. Unpredictable pedestrian behavior
2. Deficiencies in traffic infrastructure
3. Partial sensor failures
4. Objects, adverse weather, or optical conditions absent from the training data

Ex; After a nighttime crash, the crashed vehicle's lights don't work, so detection is delayed.

What is crucial is that, as long as reliance is placed on typical causal models, safety cannot be guaranteed, and this recognition must be explicitly held by the regulatory side.

The notions of "foreseeability" and "avoidability," which are questioned in medical malpractice litigation, are likely to become core points of contention in autonomous driving accidents as well. Ultimately, foreseeability equals inferring alternative causal pathways.

No.6: Formalizing the Unobservability of Purpose Hierarchies (Its Relation to Institutional Responsibility)

E2E is defined as the following mapping:

Let .

X = the space of observed states

A = the action space

θ = the learned weights

Then .

$$F\theta : X \to A$$

Next, the internal structure is defined as an inseparable composite function:

$$F\theta = H\theta \qquad \text{(i.e., } f\theta = h\theta\text{).}$$

The minimal conditions required for institutional responsibility to hold are:

・Foreseeability

・Avoidability

In causal terms, this requires:

$$\text{Estimate}\,[\text{Harm} \mid do(a')\,] < \text{Estimate}\,[\text{Harm} \mid do(a)\,]$$

That is, when some action a is selected, it must be possible to evaluate the counterfactual outcome of an unselected action a'.

Counterfactual evaluation requires causal-structure observability, and causal-structure observability requires that the following three elements be reconstructible from the outside:

・The state of the purpose hierarchy
・The conditions for purpose switching
・The causal effect of each action

In current E2E systems, what is observable and obtainable as ordinary logs consists only of the following four types:
・Input x
・Output a
・Internal activations (partial)
・Weights θ

The purpose function P(purpose state | x) does not exist as an explicit variable.
The reason is that the current purpose hierarchy is latent in θ
Expressed as an identifiability problem, in current E2E systems, for the same input x, different internal causal structures C1, C2 may exist that produce the same output:

$$f_{\theta 1}(x) = f_{\theta 2}(x) \quad \text{even though} \quad C1 \neq C2$$

This constitutes non-identifiability: the causal structure cannot be uniquely reconstructed from observational data.
Therefore, the purpose hierarchy must not remain latent; it must be externally reconstructible.

Case-1: Internal causal structure C1: Action selection based on Purpose①:
Purpose hierarchy: in the case of driving on a general road with one lane per direction
Purpose①: Compliance with traffic regulations > Minimization of accident damage based on protection of human life
Subordinate layer of the purpose hierarchy (Level A: Collision avoidance > Level B: Damage mitigation)
＊Internal reasoning:
"Deviation into the oncoming lane constitutes a violation of traffic regulations and is not permitted under Purpose①."
"Collision avoidance (①-A) is physically impossible."
"Therefore, continue braking for ①-B (damage mitigation)."
Result: Output a = forced braking.
Here, the braking is the result of prioritizing legal compliance.

Case-2: Internal causal structure C2: Action selection based on Purpose②
Purpose hierarchy: in the case of driving on a general road with one lane per direction
Purpose②: Protection of human life > Compliance with traffic regulations
Subordinate layer of the purpose hierarchy (Level A: Collision avoidance > Level B: Damage mitigation)

＊Internal reasoning:

"Collision avoidance (②-A) by braking is impossible"

"Evaluate switching from evasive braking to evasive steering."

(Switching to a deviation into the oncoming lane, a violation of traffic regulations)

＊Reason1: Because an oncoming vehicle was detected in the opposite lane, the system judged that steering-based evasion would not satisfy the higher-level purpose of protecting human life.＝Collision avoidance (②-A) is impossible

＊Reason2: No oncoming vehicle was present in the opposite lane; however, because the deceleration of the preceding vehicle is large and the TTC sharply decreases, TTC required for evasive steering is not satisfied (Time Budget > TTC), and as a result, the success probability of steering-based evasion was judged to be low.＝Insufficient steering maneuver time: The higher the ego vehicle's speed, the shorter Time Budget for evasive steering becomes, and the shorter TTC for both braking evasion and steering evasion becomes.

＊Reason1,or 2 → "Therefore, continue braking for ②-B (damage mitigation)."

Result: Output a= forced braking.

Here, the braking is the result of prioritizing protection of human life.

Institutional Problems Revealed by CASE-1 & 2

Foreseeability

"Could evasive steering have been possible?"

In Case-1: "Should the system have switched from Purpose①-A to ①-B?"

"Is the reason that evasive steering did not surface as a candidate action attributable to the system's internal structure?"

In Case-2: "Should the system have switched
from Purpose②-A(evasive braking) to ②-B(braking for Damage mitigation)?"

"Was the evaluation of the success probability of steering-based evasion appropriate?"

→ Because the purpose hierarchy is unobservable, this cannot be evaluated.

Avoidability

"Would an alternative action a' (evasive steering) have resulted in lower harm?"

And, more fundamentally, was the alternative action a' (evasive steering) even considered—given that the underlying Purpose hierarchy is itself unobservable? (In Case-1, due to the structure of Purpose①, the alternative action a' (evasive steering) may not surface as a candidate action.) → Counterfactual evaluation is impossible.

Accountability

Even if the same braking action is taken, its meaning differs depending on whether it resulted from "prioritizing legal compliance" or "prioritizing protection of human life."

→ Since this cannot be distinguished externally, responsibility attribution becomes impossible.

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Case-3: Internal Causal Structure C1:Action selection based on Purpose①

Purpose hierarchy: in the case of driving in the center-side lane (the passing lane) on a general road with two lanes per direction

Purpose①: Compliance with traffic regulations > Minimization of accident damage based on protection of human life.

Subordinate layer of the purpose hierarchy (Level A: Collision avoidance > Level B: Damage mitigation)

＊Internal reasoning:

"Because TTC < Time Budget, collision avoidance by braking (①-A) is physically impossible."

"Left steering is legally permissible; therefore, as ①-A, evasive maneuvering into the left space ahead of the leading vehicle is considered."

"A bicycle (or a motorcycle) is recognized traveling alongside on the left of the ego vehicle."

→ Left evasion judged impossible (respect for human life).

Although evasive maneuvering into the oncoming lane constitutes a traffic violation and is not permitted under Purpose①, it is judged that the risk to human life is lower than in the case of left steering. As a result, right steering is selected.

After verification that no oncoming vehicle is present: Result: Output a= right steering.

Case-4: Internal causal structure C2: Action selection based on Purpose②

Purpose hierarchy: in the case of driving in the center-side lane (the passing lane) on a general road with two lanes per direction

Purpose②: Protection of human life > Compliance with traffic regulations

Subordinate layer of the purpose hierarchy (Level A: Collision avoidance > Level B: Damage mitigation)

＊Internal reasoning:

"Collision avoidance (②-A) by braking is impossible"

"Evaluate switching from evasive braking to evasive steering."

(Switching to a deviation into the oncoming lane, checking whether an oncoming vehicle is approaching)

After verification that no oncoming vehicle is present: Result: Output a= right steering.

Institutional Problems Revealed by CASE-3 & 4

As a result, Case-3 and Case-4 generate the same output.

Case-3 constitutes, while formally adopting the purpose hierarchy (Legal Compliance > Protection of Human Life), nothing other than an internal conditional modification of the objective function: (a conditional objective function is activated.)

"If it is judged that human life cannot be protected through lawful action, then even in violation of traffic law, protection of human life shall be prioritized." Formally, the purpose hierarchy appears to be maintained; however, in substance, it has been re-ranked to (Protection of Human Life > Legal Compliance).

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What is crucial at this point is that, within an E2E-type architecture, the following are not externalized as explicitly observable variables:

・The internal representation of the purpose hierarchy
・The existence of exception rules (a conditional objective function)
・Whether the objective function has been switched
・The timing at which such switching occurred

In an E2E model, only

$$F\theta: X \rightarrow A$$

is externally observable.

The purpose hierarchy, its ranking, and any reconfiguration thereof are all latent within the weights θ.

In other words, even if the same input x produces the same output a, it becomes impossible to uniquely reconstruct from observable data:

・Whether the choice was made under a fixed purpose hierarchy;
・Whether it resulted from a conditional modification of the objective function;
・When such a modification occurred.

This demonstrates the structural problem of E2E architectures: the state-transition information of the purpose hierarchy—indispensable for determining accident responsibility—cannot be reconstructed from outside the system.

No.7: Causal Reasoning and Institutional Responsibility

The causal hierarchy introduced by Judea Pearl explains relationships among observations, interventions, and counterfactual outcomes. However, institutional responsibility attribution appears to require an additional normative layer that specifies the purpose governing action selection. Because the causal hierarchy itself does not encode such purpose structures, causal inference alone may be insufficient for institutional responsibility attribution.

This observation suggests that integrating causal reasoning with explicit purpose structures may constitute an important direction for future research on explainable and accountable autonomous systems, and clarifying this relationship between causal reasoning and institutional design remains an open problem for the safe deployment of advanced autonomous systems

No.8: Propensity Score Evaluation of E2E Autonomous Driving Models

In explainable autonomous driving systems, decision-making can be understood as understood as a hierarchical structure consisting of a higher-level objective (Purpose) and specific control actions (Task).

Based on this framework, this paper introduces the concept of the propensity score as a statistical method for comparing the behavioral characteristics of different E2E autonomous driving models.

In statistical causal inference, the propensity score is widely used as a measure representing the probability that a particular intervention is selected given observed covariates. This concept was originally proposed by Donald Rubin and has been recognized as an important method for reducing selection bias in observational data when estimating causal effects. Rubin's definition is as follows.

The probability that treatment T is assigned given the observed covariates.

$e(x)=P(T=1|X=x)$: Binary treatment propensity score

x denotes the observed covariates, i.e., background information describing the state of the system before the treatment (intervention) is selected.

Furthermore, this concept can be applied to the analysis of driving operations in autonomous driving systems.

In a virtual environment (simulation space), identical driving conditions are established, and different E2E autonomous driving models are executed along the same route. Environmental information such as road conditions, vehicle speed, surrounding traffic, and weather is represented as the observed covariate X. Driving operations such as hard braking, gentle braking, right steering, left steering, rapid acceleration, and smooth acceleration are defined as interventions.

Under this framework, the propensity score can be formulated as

$$PSe(x)=P(intvtn|X=x) \qquad \text{intvtn denotes intervention}$$

where $PSe(x)$ represents the probability that a specific driving operation is selected given the observed covariate

$$X=x$$

Using this formulation, it becomes possible to statistically compare the tendencies of different E2E models in selecting driving operations under identical environmental conditions. For example, one model-α may exhibit agile and sporty behavior characterized by frequent acceleration/deceleration and steering maneuvers, while another model-β may adopt a more conservative driving style with larger safety margins and smoother operational responses.

Here, m denotes the autonomous driving model under consideration.

Thus,

$$\text{Model}: m \in \{ E2E(\alpha), E2E(\beta) \} \qquad \alpha, \beta \text{ denote E2E model type}$$

represents the set of models being compared.

The propensity score for an autonomous driving model is defined as

$$PSe\text{-}m\ (a \mid x) = P\ m\ (A=a \mid X=x)$$

where x denotes the observed environmental state and a represents a driving action(Task). This expression represents the probability that model "m" selects action "a" when the system is in state "x".

For a given state x, the propensity score can be represented as an intervention propensity vector:

$$PSe\ (x) = \{ p1(x), p2(x), p3(x), p4(x), p5(x) \}\text{^}T$$

where each component corresponds to the probability of selecting a specific driving operation:

p1(x): hard braking
p2(x): gentle braking
p3(x): right steering including steering rate
p4(x): left steering including steering rate
p5(x): acceleration including acceleration rate

However, the following condition must hold : $\Sigma pi\ (x)=1$

Note: Although the variables are continuous in practice, they are discretized for statistical comparison.

Using the propensity score formulation, it becomes possible to quantitatively compare the behavioral characteristics of different E2E autonomous driving models (α, β) based solely on observable driving actions, without requiring direct access to their internal model structures.

Furthermore, the observed covariate X represents background information that influences both the selection of interventions (driving operations) and the resulting outcomes. Therefore, by appropriately defining the set of covariates, the behavioral characteristics of different E2E models (α, β) can be evaluated in a more objective and systematic manner.

However, this approach involves several technical issues that require careful attention. To ensure causal validity, the set of covariates must be sufficiently rich to block all backdoor paths between the environmental state and the selected driving operation. If relevant covariates are omitted, the estimated propensity scores may remain confounded, leading to biased comparisons of behavioral tendencies across E2E models. In this paper, propensity scores are not used for causal effect estimation but as a statistical representation of action-selection tendencies.

Proposals (based on the main text and Supplementary Explanations in this paper)

In recent developments, to ensure compliance with ISO 21448 and ISO 26262, the primary development axis for autonomous driving systems has been defined as:

Rule-based safety layer: Full-stack perception, decision-making, and control compliant with SOTIF / ASIL.

+E2E modules: Performance-oriented perception and decision-making, excluding control actions.

+Fail-safe monitoring: dedicated supervision and transition to a minimal-risk condition.

Within this context, the technical challenges associated with L4-level autonomous driving using rule-based approaches are generally described as follows:

＊Explosive growth of traffic scenarios

＊Necessity of handcrafted coding

＊Enormous maintenance costs

(In other words, as the ODD expands, a fundamental challenge emerges: whether complete containment within a rule-based framework is practically achievable.)

Even assuming that Fail-safe monitoring is a mandatory component, when considering the combination of E2E performance and a Rule-based safety layer, several critical challenges arise. This is particularly true when taking into account the inherent capability of vehicles to operate far faster and more agilely than human abilities, as well as rapidly changing traffic situations and high-consequence long-tail events. Under these conditions, real-time capability is essential for the selection and execution of autonomous driving maneuvers.

When the operation selected by the Rule-based safety layer as the outcome of its perception and decision-making conflicts with the operation selected by the E2E modules as the outcome of its decision-making, the following questions arise.

＊What is the division of roles between the rule-based safety layer and the E2E modules?

＊What alternative operation will be selected? (See Supplementary Explanation.No.2)

Rule-based safety layer is incapable of exploring alternative operations.

Therefore, does the Rule-based safety layer's decision-making take priority?

If so, what is the significance of the decision made by the E2E modules?

Or is it simply a reduction of the originally selected operation?

Is the E2E modules required to make a decision again?

＊Is there only one alternative, or are multiple options available?

＊How is the final operation chosen from multiple alternatives?

＊As a result, to what extent is real-time performance degraded?

I consider the essence of these challenges to lie in the broader ODD of E2E-type systems.

While the ideal objective is autonomous driving anywhere and at any time, practical

limitations emerge in the following scenarios:
* Complex and crowded urban areas
* Ambiguous construction zones

Note: Guidance gestures performed by construction-site traffic controllers, pedestrians, or cyclists toward autonomous vehicles do not, strictly speaking, entail any legally recognized responsibility. Furthermore, it is already acknowledged that a fundamental technical challenge persists concerning the extent to which autonomous vehicles can accurately perceive and interpret such guidance that is not based on standard traffic signals.

* Pedestrians that are difficult to perceive under low-light conditions
* Subtle human driver merging decisions
* Traffic signs that are difficult to recognize, for example, when positioned behind large vehicles

Considering these factors, ODD-limited, "Corridor-Type" autonomous driving appears to be an effective approach. Moreover, within the Corridor-based E2E methodology, if an operation selected by the E2E system is rejected, alternative operations can be pre-configured according to the specific characteristics of each corridor.

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The following section explains the "specific proposals" that presuppose Corridor-Type autonomous driving.

As a first step, I present a proposal to enhance explainability and regulatory compliance while maintaining the existing E2E autonomous driving architecture. This proposal does not aim to achieve full real-time causal inference, but rather to implement a structure that enables causal reconstruction after an accident while keeping computational load low.

First, it introduces an "Emergency-Only High-Density Recording Mechanism (ETCS)" that maintains a single control output during normal driving but transitions to a high-precision evaluation mode only when the Time-to-Collision (TTC) or an integrated risk metric falls below a predefined threshold. In this mode, short-term rollout evaluations of the top candidate tasks are performed, and metrics such as TTC, Time Budget, purpose state identifier (value state ID), purpose switching time, candidate task set, and evaluation distribution are recorded. This approach enables causal analysis before and after accidents without increasing constant computational load.

Note: ETCS refers Emergency Only-Time sequence-Causal information System

Additionally, probability distributions within E2E (top-k outputs and entropy) are saved as standard logs, lowering the cost of improving explainability regarding "why a particular operation (task) was selected?"

Note: top-k refers to the k most probable candidate tasks produced by the E2E model.

To prevent search space explosion, ODDs are restricted to "Corridor-type", and permissible avoidance tasks are predefined for each operating section. This structurally reduces the candidate set, balancing real-time performance and explainability.

Furthermore, value judgments are separated from the E2E weight space and managed by an external "Value State Machine." By explicitly defining states such as "Legal Compliance Priority" (Compliance with traffic regulations), "Human Life Priority" (Protection of human life), and "Damage Minimization" (Minimization of accident damage), and by establishing "transition conditions" based on TTC (Time to Collision) and Time Budget (the time required to complete the maneuver), a structure is created that enables loggable and system-integrated purpose switching.

When evaluating candidate tasks, not all branches are treated equally. Hazardous branches are prioritized using a Risk-Predictor, and detailed evaluation is performed only on the top-ranked ones. This approach ensures substantive safety assessment while reducing computational load.

Rather than performing complete counterfactual estimation in real time, high-density logs are encrypted and stored upon an accident, enabling offline counterfactual analysis in the cloud. By generating reports that compare collision probabilities and expected damage values between executed tasks and unselected tasks, this mechanism provides technical support for legal liability determinations.

Finally, maneuverability is intentionally restricted. Strategic control of maximum speed, minimum following distance, and ODD during adverse weather conditions expands the TTC (Time to Collision). Purpose switching frequently varies with the ODD and external environmental conditions—such as nighttime driving, adverse weather, or differences between urban streets and highways. In contrast, the shortening of the Time Budget (the time required to complete the maneuver) resulting from improvements in vehicle performance primarily affects task-level switching and has less influence on purpose-level switching.
Ref.: The higher the vehicle speed, the shorter the Time Budget for evasive steering becomes, and the required steering angle decreases, and the shorter the TTC for evasive braking becomes.

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The second-step evolution of autonomous driving systems should not merely aim for increased intelligence or enhanced maneuverability. Instead, it should focus on clarifying causal separation structures and strengthening regulatory compliance under bounded operational complexity.
The ultimate target structure is a hierarchical architecture with a clear separation between

the purpose layer and the task layer:

＊Purpose layer: Explicitly defines value hierarchies, such as legal compliance, prioritization of human life, and damage minimization, externalizing the objective function, associated weights, and switching rules.

＊Task layer: Generates candidate operations based on these purposes and evaluates their intervention effects. Furthermore, a limited counterfactual module enabling short-term prediction of unselected actions should be integrated incrementally.

However, merely adding causal inference capabilities within the model is insufficient to enable this structural evolution. What is fundamentally crucial is the strategic control of the system's physical and operational degrees of freedom.

When a vehicle operates at mobility and speed levels higher than those achievable by humans, the number of combinations of candidate tasks increases exponentially, and the frequency of purpose switching rises accordingly. Consequently, causal inference becomes unstable, and explainability becomes structurally challenging. As a prerequisite for long-term evolution, a strategy unique to autonomous driving that intentionally limits mobility performance in ways that ordinary human drivers could not feasibly impose on themselves, is indispensable.

Specifically, the strategy consists of the following measures:

☆Limitation of maximum vehicle speed

→ The speed limit is adjusted according to road conditions, weather, and time of day.

☆Increase in the minimum following distance

→ The distance is adjusted not only according to vehicle speed but also according to the size of the preceding vehicle and the payload mass of the ego vehicle.

☆Restriction of abrupt steering angles, braking, and acceleration

→ This requires additional vehicle capabilities, particularly effective under high-payload conditions:

＊Equipment enabling rapid lateral movement at low speeds through small yaw motion.

＊Equipment that stabilizes vehicle posture during longitudinal and lateral motion on low-μ surfaces even when small-size tires are used.

☆Reduction of the ODD during nighttime and adverse weather conditions

→ The selection of safer alternative routes according to conditions such as time of day, weather, regional characteristics (urban, suburban, and rural areas), and congestion.

By institutionally incorporating these four measures, TTC (Time to Collision) is further expanded compared to the first step, making it possible to select tasks even when their Time Budget (the time required to complete the maneuver) is large. Furthermore, the expansion of vehicle equipment can shorten the Time Budget, thereby enlarging the time margin (=

TTC−Time Budget), thereby making it easier to select the optimal task that maximizes safety value (i.e., reward) while minimizing changes in vehicle posture, thereby stabilizing Purpose switching. Following collision risk recognition, the candidate task set decreases over time, leading to a contraction of the reachable state space. Early task selection therefore suppresses the frequency of state transitions by reducing the need for repeated replanning. As a result, the combinatorial complexity associated with evaluating Purpose | Task interactions remains within a computationally tractable range. Consequently, the evaluation of Purpose | Task interactions can be performed within real-time operational constraints.

As argued in Supplementary Explanation No.3 Subsection-2, if "identical driving policy models" are to be treated as units of institutional regulation, the central obstacle lies in the structural mismatch between E2E architectures and institutional responsibility frameworks. As long as the internal causal structure of E2E remains unobservable, it cannot satisfy the legal requirements of "foreseeability" and "avoidability." (See Supplementary Explanation.No.5)

The root of this problem is that observable logs are not equivalent to the observability of causal structure.

"Observable logs ≠ Causal-structure observability"

Therefore, the issue is not merely the technical "black-box" nature of E2E, but "a deeper structural incompatibility" between E2E autonomous driving and institutional design.

When viewed in light of the overall argument developed in this paper, it appears that the following steps are required to overcome this structural incompatibility.

＊First step: In the initial stage, practical regulatory compliance is achieved by ensuring explainability through a "causally loggable E2E".

＊Second step: In the subsequent stage, without relying on computational constraints, causal explainability is structurally guaranteed through a "causal hierarchical architecture that clearly separates Purpose and Task (Purpose | Task).

After these steps are introduced, it becomes necessary to systematically evaluate how architectural modifications influence driving behavior.

In this context, the propensity score framework described in this paper (Supplementary Explanation No.8) provides a practical evaluation method.

By analyzing the probability distribution of observable driving operations (Tasks) under identical environmental conditions in a virtual environment, it becomes possible not only to quantitatively compare the behavioral tendencies of different E2E models without direct access to their internal structures, but also to evaluate how modifications to the internal model structure affect the selection of driving operations (Tasks).

Consequently, it enables empirical verification of whether architectural changes—such as the introduction of causally loggable E2E mechanisms or the structural separation of Purpose and Task (Purpose | Task)—actually lead to more consistent, explainable, and institutionally compatible driving behavior.

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In the near future, not only autonomous vehicles but also humanoid robots are expected to be socially accepted and to operate in close proximity to humans, with all of their actions controlled by AI. In that case, AI controlled humanoid robots and E2E autonomous vehicles can attain genuine social acceptability only when they are integrated into the social structure not merely as tools, but as legitimate and truly autonomous "AI-controlled agents" within human social institutions.

In this sense, I consider that not only autonomous vehicles but also humanoid robots must be equipped with a causal hierarchical architecture capable of resolving their more foundational structural incompatibility with institutional design.

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Additional Sources Consulted

Studies published in *Journal of Epidemiology* and *Japanese Journal of Biometrics* were also consulted as representative sources of causal inference research in Japan.

＊The central thesis of this paper was developed independently.

The references listed above are provided solely for contextual and conceptual background.

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Afterword

On January 15, 2009, an Airbus A320 departing from New York LaGuardia Airport struck a flock of birds shortly after takeoff, resulting in the simultaneous loss of thrust in both engines. The aircraft encountered an extremely rare (long-tail) event, dual-engine failure at a low altitude of approximately 2800 ft (850 m), and had neither the altitude required to complete the checklist nor the glide range necessary to return to an airport.

Approximately three and a half minutes (208 seconds) after takeoff, the captain executed a ditching on the Hudson River.

At the subsequent NTSB hearing, comprehensive evaluation was conducted that accounted for the actual time required for the following operational actions:

- Engine-restart attempts (APU start, re-ignition, fuel-system reconfiguration)
- Communications with ATC (status report and return-feasibility check)
- Pilot workload (airspeed management, stall prevention, maintenance of glide angle)
- Cognitive decision time (situational assessment, evaluation of options, selection of actions)

The NTSB performed counterfactual analysis (simulation) incorporating these elements and concluded that "returning to an airport was not realistically possible, and the captain's judgment and operational choices were appropriate." In the terminology of this paper, the central focus of the investigation was the counterfactual analysis regarding the validity of the decision-making: whether any possibility of returning to an airport had existed.

＊the divergence between real-time counterfactual reasoning and post hoc simulation

From this perspective, what is important is that, in emergency decision-making, counterfactual reasoning performed by the operator, whether human or AI, under conditions of limited information and time, plays an essential role, whereas post-accident counterfactual analysis (see Section-IV Subsection-6) conducted in virtual environments can have its inference results influenced by multiple bias factors, such as hindsight bias, simplified causal assumptions used to compensate for incomplete information and idealization of the sequence of actions leading up to decision-making.

This case also shows that the accuracy and depth of real-world pre-decision causal reasoning constitute foreseeability and thereby determine the avoidability of fatal accidents. Furthermore, the ability to explicitly articulate such causal reasoning constitutes explainability and, within legal-institutional frameworks, accountability.

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I affirm that I have no affiliations—direct or indirect—with any automotive corporation, semiconductor enterprise, political entity, or religious organization. For reasons of confidentiality, I respectfully request that this material be destroyed after it has been read.

With sincere respect

Akira. Revised Mar. 27 2026